

Battery Management System (BMS): Group 4 Communication & System Integration:

March 04, 2026

Our group serves as the central **communication and diagnostic hub** of the Battery Energy Management (BEM / BMS) system. It collects, validates, prioritizes, and transmits internal battery data via **CAN bus** to the Vehicle Control Unit (VCU), while hosting the UDS diagnostic server for external tools (OBD-II testers, workshops, flashing stations).

1. Internal Data Inputs that we get:

We aggregate data from three upstream groups using fast local links (SPI / I²C / shared memory / GPIO).

- **Group 1 – Monitoring** (raw sensor data)
 - Individual cell voltages (0–4.2 V, 12-bit)
 - Pack current (± 500 A via shunt)
 - Pack temperature (-20 °C to $+60$ °C)
 - Data validity flag + CRC/timestamp
 - Sensor redundancy & health status → Update rate: 100 ms (10 Hz), latency <50 ms critical
- **Group 2 – State Estimation** (calculated states)
 - State of Charge (SOC): 0–100 % (8-bit)
 - State of Health (SOH): 0–100 % (8-bit)
 - State of Power (SOP): max available power (16-bit)
 - Estimated time/energy remaining, model confidence → Update rate: 500 ms nominal (2 Hz), 100 ms for dynamic/critical SOP
- **Group 3 – Safety & Protection** (fault & reaction signals – high priority)
 - Over/undervoltage, overcurrent, overtemperature flags
 - Specific fault codes (8-bit, ISO 26262 ASIL C mapped)
 - Reaction status & safe state ready indicator → Event-driven + fast interrupt/GPIO for immediate faults

2. Bi-directional CAN Interface: with Vehicle Control Unit (VCU)

Protocol: High-speed CAN (likely 500 kbps)

Key Outputs:

- Max Charge Power / Max Discharge Power (kW, 16-bit)
- Thermal Warning (Bool)
- Precharge Command
- Battery Ready (after all safety checks)

These signals allow the VCU to enforce torque limits, charging control, thermal derating, and safe operation.

3. External Diagnostic Interface (OBD-II / UDS)

Protocol: UDS (ISO 14229) over CAN + ISO-TP (ISO 15765) Group 4 implements the full UDS Diagnostic Server (A4).

Supported Tester Commands (selected):

- Session Control (0x10) → extended diagnostic session
- ECU Reset (0x11)
- Read Data By Identifier (0x22) → live SOC, SOH, temperature, pack voltage/current
- Read DTCs (0x19)
- Clear DTCs (0x14)
- Security Access (if required for flashing/calibration)

Key Data Exposed to Tester (PIDs / DIDs):

- SOC (%), SOH (%), pack voltage (V), pack current (A)
- Battery temperature, cell imbalance (mV diff)
- Remaining capacity (kWh)
- Stored DTCs (e.g., P0AA0 – Battery Voltage Low, P0AA1 – Battery Voltage High)







